WORKED SOLUTION

STEP III 1988, Question 4

Pure • Arc Length Optimisation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q4

The key idea

Write the total length of the road as a function $L(b)$ and differentiate. The two pieces in the vertical plane must run along the surface, so their length is an arc length of the parabola; the semicircle has length πb. Differentiating an arc-length integral with respect to its lower limit needs no integration at all, and the sign of $L'(b)$ settles everything.

(The printed formula for the cross-section has $x =$ on the left; it should of course read $z =$.)

Solution

Setting up the length

Let $h(b) = (a^2 - b^2)/(2a)$, the height of the hill at distance b from the axis. The road must lie on the ground, so a path in the (x, z) plane from $(-R, 0)$ to $(-b, h(b))$ has no choice: it runs along the flat plane from $x = -R$ to $x = -a$, and then up the parabola from $x = -a$ to $x = -b$. Segment (iii) is its mirror image.

On the hill $z = (a^2 - x^2)/(2a)$, so $\dfrac{\mathrm{d}z}{\mathrm{d}x} = -\dfrac{x}{a}$, and the arc length from $x = b$ to $x = a$ is

$$\int_b^a \sqrt{1 + \frac{x^2}{a^2}}\,\mathrm{d}x.$$

The semicircle (ii) is horizontal, at height $h(b)$, with radius b (it is the circle of latitude of the hill through $(\pm b, h(b))$), so it has length πb. Hence

$$L(b) = 2(R - a) + 2\int_b^a \sqrt{1 + \frac{x^2}{a^2}}\,\mathrm{d}x + \pi b, \qquad 0 \le b \le a.$$

This also covers $b = 0$ (no semicircle; the road goes straight over the summit) and $b = a$ (the road runs round the foot of the hill).

Minimising

By the fundamental theorem of calculus, differentiating with respect to the lower limit gives minus the integrand there:

$$L'(b) = \pi - 2\sqrt{1 + \frac{b^2}{a^2}}.$$

Ask yourself Where would the stationary point be, and is it allowed?

$L'(b) = 0$ would need $1 + b^2/a^2 = \pi^2/4$, that is $b = \frac{a}{2}\sqrt{\pi^2 - 4} \approx 1.21a$. That is larger than a, so it lies outside the allowed range $0 \le b \le a$. In fact, for $0 \le b \le a$ we have

$$2\sqrt{1 + \frac{b^2}{a^2}} \le 2\sqrt{2} \approx 2.83 < \pi,$$

so $L'(b) > 0$ throughout the interval. L is strictly increasing, and its minimum is at the left end-point, $b = 0$.

Advice: take $b = 0$. The road should go straight up one side of the hill and down the other, with no semicircular detour.

How much it saves

For interest, the antiderivative is

$$\int \sqrt{1 + \frac{x^2}{a^2}}\,\mathrm{d}x = \frac{x}{2}\sqrt{1 + \frac{x^2}{a^2}} + \frac{a}{2}\sinh^{-1}\frac{x}{a} + C,$$

(substitute $x = a\sinh t$), so

$$L(0) = 2(R - a) + a(\sqrt{2} + \ln(1 + \sqrt{2})) \approx 2(R - a) + 2.296a,$$

while going round the foot gives $L(a) = 2(R - a) + \pi a \approx 2(R - a) + 3.142a$. Going over the top saves about $0.85a$.

Checks

- The answer does not depend on R, as it shouldn't: the flat stretches have total length $2(R - a)$ whatever b is.
- A rough sanity check: the hill is only $a/2$ high and $2a$ wide, so going over it cannot add much to the straight distance $2a$; the semicircle round the foot is πa. Indeed $2.296a$ lies between $2a$ and πa.
- Numerically, with $a = 1$ and $R = 3$, evaluating $L(b)$ by quadrature at 201 points of $[0, 1]$ gives the smallest value at $b = 0$; and $L'(a) = \pi - 2\sqrt{2} \approx 0.313$ is still positive.

Common slips

- Setting $L'(b) = 0$ and quoting $b = \frac{a}{2}\sqrt{\pi^2 - 4}$ without noticing that it is bigger than a.
- Using straight-line (chord) lengths for segments (i) and (iii): those would tunnel through the hill.
- Forgetting that a minimum on a closed interval can occur at an end-point.

Beyond STEP

Looking for the shortest path on a surface is the problem of *geodesics*, met in a first course on the calculus of variations or differential geometry. On a surface of revolution, Clairaut's relation describes the geodesics neatly; the king's family of roads is a small, one-parameter slice of that problem. The two analysis facts used here, differentiating an integral with respect to a limit and the fact that a continuous function on a closed interval attains its minimum, are proved carefully in Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.